KIRAN SHIVAKUMAR

Application Engineer · LabVIEW · Robotics

CONTACT

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CORE SKILLS

- LabVIEW
- Test & Measurement / DAQ
- Python
- Computer Vision
- ROS / Robotics

LABVIEW ARCHITECTURE

Queues · Notifiers · State Machine Architecture · JKI State Machine · Queued Message Handler (QMH) · Delacor QMH (DQMH)

AI & VISION

OpenCV · TensorFlow · MediaPipe · YOLO v5/v8/v11/26 · DeepSORT

ROBOTICS PLATFORM

ROS · Gazebo · RViz · ROS SLAM

EDUCATION

B.E. in Mechatronics

The Oxford College of Engineering

2017 – 2021 ·

8.1 CGPA

PROFESSIONAL PROFILE

Dedicated Application Engineer with **5+ years** of experience designing and deploying production-grade **Automated Test Equipment (ATE)**, Hardware Abstraction Layers (HAL), and condition monitoring across aerospace, defense, space, and energy sectors. Expert in multi-vendor instrument control (SCPI, VISA, Modbus), high-speed sensor DAQ, and integrating **Python, Computer Vision (YOLO/OpenCV), and ROS** to automate quality qualification and cut test cycle times.

5+	7+	15+	11
Years Experience	Client Projects	Instruments	Protocols

WORK EXPERIENCE

Application Engineer

Feb 2024 – Present

Ducom Aerospace Pvt Ltd · Bengaluru, India

- Architect **LabVIEW & Python ATE/DAQ systems** for tribology test benches, acquiring multi-channel sensor data (force, temperature, vibration, acoustic emission) at up to **10 kHz**.
- Built an automated **wear scar optical measurement system** using custom **YOLOv5**, PyQt5 daemon, and LabVIEW TCP sequencing—cutting inspection cycle time by **85%**.
- Architected modular applications utilizing **Queued Message Handler (QMH)**, **Delacor QMH (DQMH)**, and **JKI State Machines** with **Queues** and **Notifiers** for deterministic asynchronous DAQ.
- Developed autonomous platform mapping, obstacle detection, and live telemetry HUD using **ROS, Gazebo, and SLAM**.

Tech: LabVIEW (QMH, DQMH, JKI), Python, OpenCV, TensorFlow, MediaPipe, YOLO, DeepSORT, ROS, Gazebo, RViz, SLAM

LabVIEW Developer

Sep 2021 – Jan 2024

Access Automation Pvt Ltd · Bengaluru, India

- Delivered **7+ turnkey automated test & DAQ systems** for premier clients including **ISRO (SAC), NPCIL, Tata Power, and LG India** with 100% on-time qualification sign-offs.
- Engineered modular **Hardware Abstraction Layers (HAL)** for SCPI/VISA, GPIB, and Modbus RTU/TCP across Keysight, Yokogawa, HIOKI, and Anritsu instruments.
- Built automated test sequences with real-time parametric limit compliance, SQLite datalogging, and automated compliance certificates (HTML/CSV).
- Interfaced laboratory instrumentation with Mitsubishi FX5U PLCs and Advantech ADAM modules for safety interlocks.

Tech: LabVIEW, ATE, DAQ, SCPI, GPIB, Modbus, RS-485, PLC, Report Generation

FEATURED CLIENT PROJECTS

DAQ & Reporting Software for Antenna Testing

SAC · ISRO

Configures antenna testing parameters (frequency, bandwidth, measurement specs), controls DUT/feed positioners, and measures radiation patterns.

HW: Keysight MXG N5183B, PNA-X N5264B, Power Meter, Spectrum Analyzer E4446A, Position Controllers

Transformer Testing Bench and Software

TATA POWER

Automated test bench executing no-load, full-load, temperature rise, vector group, DVDF, high-voltage, turn-ratio, and insulation resistance tests.

HW: Yokogawa WT500, Motwani XTRM-3A, Motwani XWRM-25A, Motwani 5kV IR Tester, FX5U PLC

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INSTRUMENTS INTEGRATED

- Keysight PNA-X N5264B
- Keysight MXG N5183B
- Keysight 33210A AWG
- Keysight DSO-X 2012A
- Spectrum Analyzer E4446A
- Yokogawa WT500 Analyzer
- Yokogawa GM10 (19 Units)
- HIOKI PW3335 Power Meter
- Anritsu MT9085 OTDR
- EXFO FTB-500 OTDR
- Advantech ADAM 4050/4060
- Motwani Turn-Ratio XTRM-3A
- Motwani Winding Res. XWRM-25A
- Motwani 5kV Diagnostic Tester
- Mitsubishi FX5U PLC

COMMUNICATION PROTOCOLS

- SCPI Commands
- PyVISA / HAL
- GPIB Interface
- TCP/IP Socket
- UDP Socket
- RS-485 Serial
- RS-232 Serial
- Modbus Protocol
- CAN Bus
- HART Protocol
- Telnet
- FTP
- USB DAQ

INDUSTRY DOMAINS

- Aerospace & Defense
- Space & RF Testing
- Nuclear Safety Monitoring
- Power & Transformers
- Fiber Optics Telecom
- Consumer Appliances
- Tribology DAQ
- Autonomous Robotics

LANGUAGES

- English
- Hindi
- Kannada
- Tamil

OPERATING SYSTEMS

- Windows
- Linux (Ubuntu)

FEATURED CLIENT PROJECTS (CONTD.)

Perimeter Automated Detection System
NPCIL · KAKRAPAR ATOMIC POWER

Monitors and acknowledges safety alarms triggered by microwave detectors and PIR perimeter sensors across nuclear plant zones.

HW: Advantech ADAM-4050, Advantech ADAM-4060 Modules

Power Consumption Test Automated Software
LG INDIA

Automated test sequence acquiring electrical parameters from power analyzers and proximity sensors with automatic pass/fail reporting.

HW: HIOKI PW3335 Power Meter, Mitsubishi FX5U PLC, Proximity Sensors

Cell Voltage Monitoring System (19 Networked Units)
GACL INDIA

Networked DAQ architecture interfacing 19 Yokogawa GM10 units across a shared industrial LAN for continuous remote cell monitoring.

HW: Yokogawa GM10 Data Acquisition System (19 Nodes), TCP/IP

ATE Software for Frequency Response
ELECTROHMS INDIA

Automated control of arbitrary waveform generator and oscilloscope to sweep frequencies, compute RMS values, and plot response in dB.

HW: Keysight 33210A AWG, Keysight DSO-X 2012A Oscilloscope

ConnectOTDR Fiber Optic Testing System
OMAN FIBER OPTICS

Multi-instrument optical testing suite evaluating wavelength, resolution, pulse width, and fault distance across fiber runs.

HW: EXFO FTB-500 Platform, Anritsu MT9085 OTDR

FEATURED OPEN-SOURCE PROJECTS

OpenATE-Engine
Enterprise Python ATE framework with SCPI/VISA Hardware Abstraction (HAL), parametric limits, SQLite logs & HTML reports.

github.com/kiran10299/OpenATE-Engine

EdgeVibe-Pro
Industrial multi-sensor edge vibration DAQ with zero-copy ring buffer, Crest/Kurtosis DSP, and Modbus TCP gateway.

github.com/kiran10299/EdgeVibe-Pro

LabVIEW ↔ Python AI Bridge
Hybrid architecture streaming LabVIEW DAQ sensor arrays into Python ML for real-time predictive anomaly detection.

github.com/kiran10299/LabVIEW-Python-AI-Bridge

Straw-Hat Reminders
Procedural vector animated companion in PyQt5 with Windows API ctypes idle tracking for active ergonomic screen-time health.

github.com/kiran10299/Straw-Hat-Reminders